ELEG 5633 Detection and Estimation
Detection Theory II

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March 18, 2015

Motivation

So far, we have learned

- Bayesian Detector: given $p(x|H_0)$, $p(x|H_1)$, π_0, π_1, cost assignment
  LRT; MAP; ML
- NP-detector: given $p(x; H_0)$, $p(x; H_1)$
  LRT
- Deterministic signal
  matched filter; generalized matched filter
- Random signal
  energy detector, estimator-correlator.

The distribution of X under H_0, H_1 are known.

What if the PDF of X is not completely known?

Composite Hypothesis Testing

- Assume the PDF of X has unknown parameters under H_0 and H_1
  $p(x|\theta_0, H_0)$ or $p(x; \theta_0, H_0)$ under H_0;
  $p(x|\theta_1, H_1)$ or $p(x; \theta_1, H_1)$ under H_1.
- Three main approaches
  - Uniformly Most Powerful (UMP) test
  - Bayesian approach
  - Generalized likelihood ratio test (GLRT)

Example

Given N observations, $x[n], n = 0, 1, \ldots, N-1$, which are i.i.d and, depending on the hypothesis, generated according to

$$H_0 : X[n] = W[n]$$
$$H_1 : X[n] = A + W[n]$$

where $W[n] \sim \mathcal{N}(0, \sigma^2)$. The value of A is unknown, although a priori we known $A > 0$.

The NP test is to decide H_1 if

$$\frac{p(\mathbf{x}; A, H_1)}{p(\mathbf{x}; H_0)} = \frac{\frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=1}^{N-1}(x[n]-A)^2\right]}{\frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=1}^{N-1}x[n]^2\right]} > \gamma$$

i.e.,

$$A\sum_{n=1}^{N-1} x[n] > \sigma^2 \ln\gamma + \frac{NA^2}{2}$$

Since $A > 0$, we have

$$T(\mathbf{x}) = \frac{1}{N}\sum_{n=1}^{N-1} x[n] > \frac{\sigma^2}{NA}\ln\gamma + \frac{A}{2} = \gamma'$$

Question: Can we implement this detector without knowing the exact value of A?

$$P_{FA} = \mathbb{P}[T(\mathbf{x}) > \gamma'; H_0] = Q\left(\frac{\gamma'}{\sqrt{\sigma^2/N}}\right)$$

So that

$$\gamma' = \sqrt{\sigma^2/N}Q^{-1}\left(P_{FA}\right)$$

For a given P_{FA}, the threshold of the NP test is independent of A!
Under H_1, $T(\mathbf{x}) \sim \mathcal{N}(A, \sigma^2/N)$, thus

$$P_D = Q\left(\frac{\gamma' - A}{\sqrt{\sigma^2/N}}\right) = Q\left(Q^{-1}\left(P_{FA}\right) - \sqrt{\frac{NA^2}{\sigma^2}}\right)$$

The NP detector yields the highest P_D for any value of A, as long as $A > 0$.

Uniformly Most Powerful (UMP) Test

Consider a composite hypothesis test $H_0 : \theta = \theta_0$ versus $H_1 : \theta \in \Theta_1$.

Definition

A UMP test is a hypothesis test that has the greatest power (i.e., P_D) $1 - \beta$ among all possible tests of a given size (i.e., P_{FA}) α for all $\theta \in \Theta_1$.

- UMP may not exist.
- For a UMP test to exist, the parameter test must be one-sided.

Consider a two sided test

$$H_0 : A = 0$$
$$H_1 : A \neq 0$$

If $A > 0$, $T(\mathbf{x}) > \sqrt{\sigma^2/N} Q^{-1}(P_{FA})$;
If $A < 0$, $T(\mathbf{x}) < -\sqrt{\sigma^2/N} Q^{-1}(P_{FA})$
Since the value of A is unknown, the NP approach does not result in a unique test.

Suboptimal Tests

When a UMP does not exist, we need to implement suboptimal tests.

- Consider the unknown parameters as realizations of random variables and assign a prior PDF.
  $\Rightarrow$ Bayesian approach
- Estimate the unknown parameters for use in a likelihood ratio test.
  $\Rightarrow$ Generalized likelihood ratio test (GLRT)

Bayesian Approach

- The Bayesian approach assigns prior PDFs to θ_0 and θ_1, denoted by $p(\theta_0)$ and $p(\theta_1)$.

$$p(\mathbf{x}; H_0) = \int p(\mathbf{x}|\theta_0; H_0) p(\theta_0) d\theta_0$$
$$p(\mathbf{x}; H_1) = \int p(\mathbf{x}|\theta_1; H_1) p(\theta_1) d\theta_1$$

- The optimal NP detector decides H_1 if

$$\frac{p(\mathbf{x}; H_1)}{p(\mathbf{x}; H_0)} = \frac{\int p(\mathbf{x}|\theta_1; H_1) p(\theta_1) d\theta_1}{\int p(\mathbf{x}|\theta_0; H_0) p(\theta_0) d\theta_0} > \gamma$$

- It requires multi-dimensional integrations.
- The choice of the prior PDFs can be difficult.

Generalized Likelihood Ratio Test (GLRT)

- The GLRT decides H_1 if

$$L_G(\mathbf{x}) = \frac{\max_{\theta_1 \in \Theta_1} p(\mathbf{x}; \theta_1, H_1)}{\max_{\theta_0 \in \Theta_0} p(\mathbf{x}; \theta_0, H_0)} > \gamma$$

- The first step is to find the MLEs of θ_0 and θ_1.

Example

Consider $X[n] = A + W[n]$, $n = 0, 1, \ldots, N-1$, where $W[n]$ is WGN with PDF $\mathcal{N}(0, \sigma^2)$. We have two hypotheses

$$H_0 : A = 0$$
$$H_1 : A \neq 0$$

What is the GLRT?

The GLRT decides H_1 if

$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{A}, H_1)}{p(\mathbf{x}; H_0)} > \gamma$$

where $\hat{A} = \arg\max_{A \neq 0} p(\mathbf{x}; A, H_1)$.